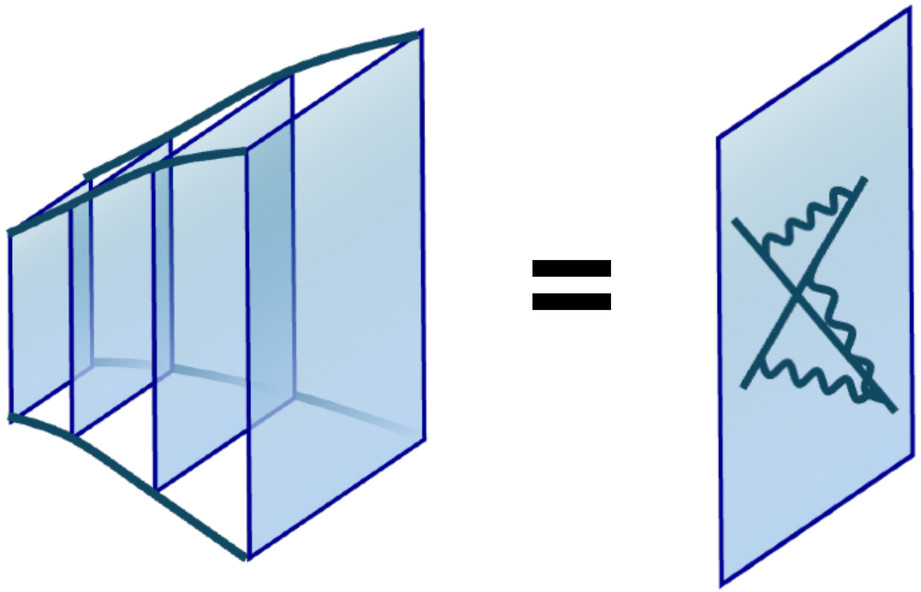
Random interlude: AdS/CFT correspondence



$$ds^2 = \frac{1}{z^2}(dz^2 + dx^2 + dy^2)$$

- The AdS_{d+1} / CFT_d correspondence from string theory tells us that sometimes theories of particles in d-dimensions are exactly dual to theories of quantum gravity living in Anti de Sitter space AdS_{d+1}.
- This **holographic duality** is a deep and profound fact about quantum gravitational dynamics.
- The most basic kinematics of this is that the **conformal group** in d dimensions is the same as the **isometry group** of AdS_{d+1}.
- More precisely: AdS_{d+1} has a $\mathbb{R}^d$ boundary, and its isometry group acts as the conformal group on points at the boundary.

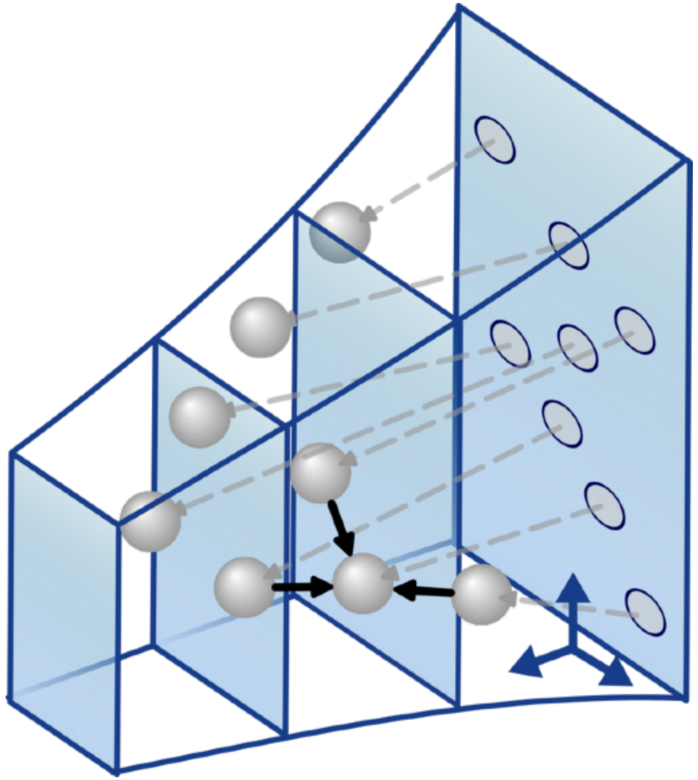
AdS-GNN

We introduce **AdS-GNN**: a **graph neural network** that uses AdS space to organize the computation.

Idea:

1. Take an input point cloud on $\mathbb{R}^2$ to be on the boundary of AdS.

2. **Lift the data** into the interior of AdS in a conformally covariant way.

3. **Do bulk operations** in a way that respects AdS isometries using known equivariant neural network tools generalized to AdS.

4. Read out the answer at the end.